

Zoe Tian

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WORK EXPERIENCE

- Meta** Bellevue, WA
Software Engineer (MRS - Modern Recommendation Systems) April 2025 - Present
 - Led cross-team technical design of an **agent-driven GNN ranking policy optimization framework** for Facebook Reels and Deep Dive surface, enabling autonomous policy search across user cohorts and ranking configurations, increasing **video view seconds (VVS)** by **6%** and **like-sum engagement** by **6.37%**
 - Owned the end-to-end policy evaluation workflow across 10+ candidate ranking strategies, including seed experiment selection, policy generation, offline metric analysis, and cross-functional launch coordination with ranking, GNN, and video recommendation teams
 - Built a full-stack prototype for causal video recommendation, designing scalable candidate retrieval pipelines and contextual recommendation interfaces using **React, TypeScript, Python, and Hack**
 - Optimized distributed data preprocessing pipeline for the ranking system, reducing CPU label parsing time by **86% (550ms to 75ms)** and machine resource usage by **25%** through UDF optimization and elimination of redundant computation
 - Mentored junior engineers and new hires through knowledge sharing, and guidance on team processes and best practice
- Dropbox** Vancouver, BC
Infrastructure Software Engineer (Search Retrieval & ML team) March 2022 - March 2024
 - Designed and implemented Daggoo: a 1.5-year project that transform query serving paths into a DAG structure in **Golang**
 - Reduced **P95 Search latency by 200ms** through the application and deployment of Daggoo
 - Set up experiments to safely verify the latency gain in each DAG-ified components with live shadow traffic
 - Created a visualization tool in **D3.js** for debugging and troubleshooting DAG execution latency
 - Repeatedly analyzed critical path with Hive and DAG logs, modified framework and call paths accordingly, until hitting latency goal
 - Improved paid enterprise user experience by fixing bugs in search with the eastern locales, and search by tags in Japanese during on-call rotations
- VMware** Toronto, ON
Member of Technical Staff II (Tanzu Application Platform) April 2020 - March 2022
 - Led the integration of Backstage.io as the developer portal foundation, reducing new service onboarding time by 70%
 - Built automated service templates in Backstage.io, reducing microservice creation time by 60%
 - Created Backstage plugins for VMware tools integration, serving 1000+ services
 - Onboarded two new members onto the team, mentored junior engineers, and facilitate internal meetings
- Pivotal Software** Toronto, ON
Software Engineer (Concourse CI team) June 2019 - April 2020
 - Maintained and developed new features for Concourse CI in **Golang and Elm** followed strict TDD and Agile
 - Scaled up the previous backend authentication flow for super admin users with large pipelines
 - Re-wrote dashboard endpoints in Elm and improved the performance by 50% when loading complex responses
 - Worked in a multi-cloud environment with **AWS, GCP, VSphere**; developed with **Docker, Kubernetes, Terraform**

EDUCATION

- University of Waterloo** Waterloo, ON
Bachelor of Computer Science, Minor in Combinatorics & Optimization 2015 - 2019

PROJECTS

- Midi Melody** github.com/zoetian/midi_melody
YouTube Music Video Converter in Python
 - Applied Youtube-dl library to convert random audio files into mp3
 - Extract the melody from an audio file and convert it to MIDI notes
 - Output piano keyboard canvas animation as learning tutorials with sheet script pdf
- Docker Filter** github.com/zoetian/docker-filter
Tiny Image Processing Experiment
 - Built a Serverless function to do style transfers with Docker, Openfaas, Tensorflow
 - Supported 40+ different styles from Magenta's models
- Bustub - CMU Database 445/645** github.com/zoetian/bustub
A relational database management system
 - Built a disk-oriented database management system in C++ to support basic SQL